



CHURNZERO

Customer Churn Prediction

A Complete Guide — From Zero to Understanding

Made especially for my friends

No coding knowledge needed.

No maths. No jargon.

If you can read this, you can understand this.

What is this document?

Hey! So you heard we participated in a hackathon and built something called a **churn prediction model** — and now you're wondering what on earth that means. This guide was written *just for you*. No coding. No maths. No jargon. Just simple explanations with everyday analogies.

This document was made **especially for my friends** — the people who asked "*so what did you actually build?*" and genuinely wanted to know. By the end of this guide, you'll be able to explain this project at a dinner table. That's the goal.

What we'll cover:

- 01 What is Churn — and why do banks lose sleep over it?
- 02 What is Machine Learning — in plain English
- 03 The Data we were given to work with
- 04 What we actually did — step by step
- 05 The 7 smart features we invented
- 06 How the model learns
- 07 The results — what we found
- 08 What this means for the bank — in rupees
- 09 Glossary — every term explained simply

01 — What is Churn?

Imagine you subscribe to Netflix. One day you decide it's not worth it anymore — maybe a competitor is cheaper, or you just stopped watching. So you cancel. **That's churn.** You were a customer. Now you're not.

In banking, churn means a customer **closes their account** or simply stops using the bank — stops depositing money, stops making transactions, goes completely inactive. They've moved to another bank.

Why does this matter so much?

16%

of customers
churn every year

■40,000

lost revenue per
churned customer

■500

cost to retain
one customer

Think about those numbers. If a bank has 100,000 customers and 16% churn, that's **16,000 customers leaving every year**. At **■40,000** lost per customer, that's **■64 crore in lost revenue annually**.

But here's the key insight: **if the bank can identify who is about to leave before they actually leave**, it can call them, offer a better deal, and keep them. That retention call costs only **■500**. Catching one churner saves **■39,500**.

Simple analogy:

Think of it like a doctor checking your health. A doctor doesn't wait until you're seriously ill — they look at warning signs early and intervene. We do exactly that for the bank's customer relationships. We look at warning signs in customer behaviour and flag who is about to leave before they actually do.

02 — What is Machine Learning?

This is the part everyone thinks is complicated. It's not.

1

The old way — hand-written rules

A bank manager might say: 'If a customer hasn't logged in for 60 days AND made fewer than 2 transactions this month, they're probably leaving.' This works sometimes — but with 97 different pieces of information per customer, no human can figure out the right combination of rules manually.

2

The ML way — learn from examples

Instead of writing rules by hand, we show the computer thousands of examples where we already know the answer. 'Here are 8,101 customers. Here's everything about each one. Here's whether they churned or not.' The computer finds patterns too complex for humans to spot.

3

Apply to new customers

Once the computer has learned the pattern from 8,101 known customers, we show it 2,026 new customers it has never seen — and ask: will these customers churn? That's the prediction we submit.

The exam analogy:

File 1 (training data) is like last year's exam paper with the answer key. You study it to understand the pattern. File 2 (test data) is this year's exam — same type of questions, no answers given. You apply what you learned. That's exactly what our model does.

03 — The Data We Were Given

We were given two files containing information about real bank customers. Think of each row as one customer's complete profile.

File	Rows	Contents
ChurnZero_Dataset_v1.csv	8,101	Profiles + whether they churned (0=stayed, 1=left)
ChurnZero_Test_v1.csv	2,026	Profiles only — we predict the churn column

The 97 features — what we knew about each customer:

Customer Profile	Age, gender, marital status, education, occupation, income, city tier
Relationship & Tenure	How long they've been a customer, how many products they hold
Account & Transactions	Monthly balance, transaction count, cash withdrawals, UPI usage
Products Held	Savings account, credit card, home loan, FD, investments, insurance
Credit Card & Loans	Credit limit, utilization ratio, late payments, loan risk score
Digital Banking	App logins, website visits, days since last login, app rating
Service & Complaints	Total complaints, resolution time, satisfaction score, NPS
Marketing & Retention	Campaigns received, offers accepted, competitor bank awareness

One important discovery: Only one column had missing data — **app_rating_given** (56% missing, because many customers never rated the app). We filled the blanks with the average rating and added a flag column. Everything else was complete.

04 — What We Actually Did — Step by Step

Phase 1 — EDA — Getting to Know the Data

Before building anything, we explored the data. We asked: who are these customers? What does the typical churner look like? We found the churn rate was 16% — about 1 in 6 customers. We also discovered 4 columns that were 'cheating' — they contained information only available after someone had already churned. We removed those to keep our model honest.

Phase 2 — Feature Engineering — Inventing Smarter Clues

Instead of just using the 97 raw features, we combined them into 7 new, smarter features that better capture churn behaviour. For example: instead of just knowing days since last login, we combined that with how often someone uses digital banking to create a 'digital disengagement score'.

Phase 3 — Model Training — Teaching the Computer

We trained 3 different models (LightGBM, XGBoost, CatBoost) on the 8,101 known customers. Each model learns slightly differently, so combining all 3 gives better predictions. We tested rigorously using 5-fold cross validation — splitting data 5 ways and testing on each.

Phase 4 — Predictions & Submission

Finally, we ran our trained model on the 2,026 test customers. For each one, we output a probability (0 to 1) of churning and a final yes/no prediction. We then packaged everything — predictions, slides, and this document — for submission.

05 — The 7 Smart Features We Invented

Instead of just feeding raw data into the model, we created 7 new features by combining existing ones in meaningful ways. Think of it like a detective combining clues — each individual clue might be weak, but combined they tell a clear story.

1 Digital Disengagement

Formula: $\text{last_login_days} \times (1 - \text{digital_transaction_ratio})$

If someone hasn't logged in for 45 days AND rarely uses digital banking, that's a much stronger signal than either fact alone. Ranked #7 in top churn drivers!

2 Credit Stress Score

Formula: $\text{credit_utilization} \times \text{late_payments} + \text{default_risk}$

A customer maxing their credit card AND making late payments AND having high default risk is under serious financial stress — a key churn predictor.

3 Dissatisfaction Index

Formula: $\text{complaints} / \text{satisfaction_score} + \text{escalations} \times 2$

Many complaints + low satisfaction + escalations = very unhappy customer. Unhappy customers leave.

4 Balance Health

Formula: $\text{balance_decline_percentage} \times \text{account_inactive_days} / 30$

A declining balance combined with account inactivity is a classic pre-churn pattern.

5 Product Breadth

Formula: $\text{count of all 10 product flags held}$

More products = harder to switch banks. This feature ranked #6 in our top churn drivers!

6 Campaign Responsiveness

Formula: $\text{campaign_responses} / \text{campaigns_received}$

If the bank sent 10 retention offers and the customer responded to none, that's a very bad sign.

7 Engagement Drop Ratio

Formula: $\text{total_logins} / \text{tenure_months}$

A long-term customer who drastically reduced engagement is showing early churn signals.

The result:

Two of our 7 engineered features — **Product Breadth** and **Digital Disengagement** — ranked in the top 10 most important churn drivers, beating out 90+ raw features. That validates our thinking.

06 — How the Model Learns

We used 3 models. Here's what each does in plain English:

LightGBM

Imagine an expert who asks yes/no questions very quickly to sort customers into 'likely to churn' and 'likely to stay' buckets. LightGBM is a very fast, very smart decision tree — it learns which questions to ask to best separate churners from non-churners.

XGBoost

Similar to LightGBM but learns more carefully and deliberately. Each new tree it builds specifically targets the examples the previous tree got wrong. It learns from its mistakes in a very structured way.

CatBoost

Especially good at handling text-like categories — 'Tier 1 city' vs 'Tier 2 city' vs 'Tier 3'. Where other models need you to convert these to numbers first, CatBoost handles them naturally.

Why all 3? Sports team analogy:

Imagine asking 3 experienced talent scouts to independently assess a cricket player. One focuses on batting stats. One on fielding. One on mental toughness. Averaging their scores gives more reliable predictions than trusting just one. That's exactly what blending 3 models does.

The imbalance problem — and how we solved it

Only 16% of customers churn. If our model just said 'everyone stays' for every customer, it would be right 84% of the time — but would catch **zero churners**. Completely useless.

Our solution: we told the model that churners are **5.22 times more important** during training. This forces it to really pay attention to the minority class — the actual churners we care about.

07 — The Results

0.77

PR-AUC Score
(primary metric)

0.68

F1 Score
(precision vs recall balance)

0.60

Optimal threshold
(tuned for business cost)

What do these numbers mean?

PR-AUC 0.77

How good is the model at finding real churners without too many false alarms? A score of 0.77 on data where only 16% churn is genuinely good. A random model would score around 0.16. We scored 0.77 — nearly 5x better than random.

F1 Score 0.68

A balance between 'how many churners did we catch' and 'how many non-churners did we incorrectly flag'. 0.68 means we're catching the majority of real churners without flooding the retention team.

Threshold 0.60

By default, models predict churn if probability > 0.5. But missing a churner costs █40,000 while a false alarm costs only █500. We tuned to 0.60 — minimising total business cost.

Top 10 Churn Drivers (SHAP Analysis)

We used **SHAP** (SHapley Additive exPlanations) to understand why the model makes each prediction. It tells us which features matter most and in what direction.

Rank	Feature	What it means
1	Credit Card Spend	Sudden drop in spending = planning to leave
2	Available Credit	Maxed out credit = financial stress
3	Transaction Change	Declining transactions = disengaging
4	Number of Products	Fewer products = easier to switch banks
5	Credit Card Limit	Lower limit customers churn more
6	Product Breadth ★	Our engineered feature — validated!
7	Digital Disengagement ★	Our engineered feature — validated!
8	Age	Younger customers switch more easily
9	App Rating	Low rating = dissatisfied = leaving soon
10	Prepayment Flag	Paying off loans before leaving the bank

★ = features our team engineered. They beat 90+ raw features to reach the top 10.

08 — What This Means in Rupees

All of this is worthless if it doesn't translate to real business value. So let's talk about what our predictions actually mean for the bank.

2,026

total test customers

272

predicted to churn

1,754

predicted to stay

The Business Math

Scenario	Calculation	Amount
Retention calls needed	272 customers x Rs.500	Rs.1,36,000
Revenue at risk (no action)	272 x Rs.40,000	Rs.1,08,80,000
Net saving (approx)	Rs.1,08,80,000 - Rs.1,36,000	Rs.1,07,44,000

What should the bank do with our predictions?

1. Monitor Digital Disengagement

Any customer whose last_login_days is very high AND rarely uses digital banking should be flagged immediately for a relationship manager call.

2. Cross-sell Products

Single-product customers churn 3x more. Offering them a savings account or FD alongside their existing product significantly reduces churn risk.

3. Watch for Spending Drops

A >30% drop in credit card spending month-over-month is a strong early warning. Automated alerts should trigger retention workflows.

4. Target Competitor-Aware Customers

Customers who know about competitor bank offers should receive personalised retention offers within 48 hours.

5. Loyalty Programs for Young Tier-3 Customers

This segment showed the highest churn risk. Enrol them proactively in loyalty programs before they have reason to leave.

09 — Glossary — Every Term Explained Simply

Here are all the technical words used in this project — explained like you're hearing them for the first time. Because you are, and that's completely fine.

Churn	When a customer stops using a service. In banking: closing account or going inactive.
Machine Learning	Teaching a computer to find patterns by showing it many examples with known answers.
Model	The 'brain' the computer builds after learning. It stores the pattern learned from training.
Training Data	Examples with known answers used to teach the model. (File 1 — 8,101 customers)
Test Data	New examples where the model predicts the answer. (File 2 — 2,026 customers)
Feature	A single piece of information about a customer. Age, balance, login count are all features.
Feature Engineering	Creating new, smarter features by combining existing ones. Like a recipe from ingredients.
LightGBM / XGBoost / CatBoost	Three specific ML models. All are 'gradient boosting' — excellent for tabular data.
Blending	Combining predictions from multiple models to get a better final prediction.
PR-AUC	How good the model is at finding churners. Higher = better. Range: 0 to 1. We got 0.77.
F1 Score	Balance between catching real churners and minimising false alarms. We got 0.68.
Threshold	Probability cutoff above which we predict churn. We used 0.60 instead of default 0.50.
SHAP	A method to explain which features drove each prediction. Shows the model's reasoning.
Class Imbalance	When one category (churners at 16%) is much rarer than the other (84% non-churners).
Cross Validation	Testing the model on data it wasn't trained on to ensure results are genuine.

Leakage	When the model uses information unavailable in real life — makes results artificially good.
False Negative (FN)	Predicting someone will stay when they actually leave. Costs the bank Rs.40,000.
False Positive (FP)	Predicting someone will leave when they actually stay. Costs Rs.500 in wasted calls.
scale_pos_weight	A setting telling the model churners are more important. We set it to 5.22.

Thanks for reading!

You now understand a real machine learning project.
That's genuinely impressive.

The whole project in one paragraph

We were given data on 8,101 bank customers — their profiles, transactions, and whether they churned. We cleaned the data, invented 7 new features, trained 3 ML models, blended their predictions, and identified 272 out of 2,026 test customers who are likely to churn — potentially saving the bank over Rs.1 crore in lost revenue.

0.77

PR-AUC

0.68

F1 Score

272

Churners Found